5.2 Properties of Rational Exponents and Radicals

Name _____ Period ____ Date _____

Algebra II 5.2 (30-minute plan) · Student sheet

I can…

- simplify expressions with rational exponents.
- write radical expressions in simplest form.
- add, subtract, multiply, and divide radical expressions.

Vocabulary · fill in the missing word

Simplest form of a radical — no perfect nth-power factors in the radicand, no fractions in the radicand, and no radicals in a _____

Rationalizing the denominator — _____ a radical from the denominator

Conjugates — _____

Like radicals — radicals with the same _____ and the same radicand

Part 1 · Notes and examples (fill in each note, then work the example)

Properties of rational exponents: $a^m \cdot a^n =$ _____ $(a^m)^n =$ _____ $a^m/a^n =$ _____

Example 1

Simplify each expression.

a. $5^{\frac{1}{3}} \cdot 5^{\frac{5}{3}}$ b. $4^{\frac{5}{2}{4}^{\frac{1}{2}}}$ c. $(3^{\frac{2}{3}})^3$

Properties of radicals: $\sqrt[n]{a \cdot b} =$ _____ $\sqrt[n]{a/b} =$ _____

Simplest form: A radical is in **simplest form** when the radicand has no perfect nth-power _____, there is no _____ in the radicand, and there is no radical in the _____.

Example 2

Write each expression in simplest form.

a. $\sqrt[5]{64}$ b. $\sqrt[4]{\frac{5}{4}}$

Conjugates: The **conjugate** of $a + b\sqrt{c}$ is _____.

Like radicals have the same _____ and the same _____.

Example 3

Write $\frac{1}{5+\sqrt{3}}$ in simplest form.

Part 2 · Partner practice

Partner practice

Write $\sqrt[3]{250}$ in simplest form.